Daniel Hardesty Lewis

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

Experience

Professional

Apr. 2025 – Present **Co-founder**, *Homecastr*, New York, NY

Targetting the last blue ocean in consumer real estate technology: individual home price forecasting

- Here's our pitch
- Built out a current home price prediction ML model, achieving 12% MAPE against Zillow's 8.4% in the hardest US market to do it: Manhattan
- That model stack is identical to some of the recent cutting-edge physics foundation models, using a Variational AutoEncoder (VAE) approach
- Uses a data-driven approach to auto-configure the vast majority of hyperparameters and even some of the model architecture, trains in under 7 min
- Interest from a16z & MetaProp

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

Top-5% YC entrant, KPMG rolled out their Tariff Modeller after we put polibom.com in front of them

- Now targetting early stage investment

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

Developed the highest quality seamless elevation data for Texas, summitgeospatial.com

- Now targetting recent interest received to license to AI labs such as OpenAI & Cohere

Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Spun out Summit Geospatial alongside UT's VPs of IP
- Led a principal project of the $40M Texas Disaster Information System (TDIS) disaster resiliency initiative
- Contributed research towards Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors
- Scaled climate and flood models on world-leading supercomputers, utilizing millions of dollars in computational resources and million-node jobs
- Evaluated and accepted tens of millions of dollars in compute requests
- Managed relationships and developed projects with federal, state agencies, and universities including NOAA, NSF, US Army Corps of Engineers, Texas General Land Office, USC, UT, A&M

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers program, focusing on disaster resiliency in East Africa
- Research received a mention from President of AAAI Yolanda Gil during her farewell speech
- Helped teach Petrobras deep learning on HPC

Research

May 2025 – Present **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY

Targetting theory publications to DeepSeek's GRPO algorithm

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dhardestylewis • *dhl*

researchgate.net/profile/Daniel-Hardesty-Lewis

Jan. 2025 – Present **Research assistant to Dir. Financial Engineering Ali Hirsa**, *Columbia University*, New York, NY

Achieved better than commercial R^2 on backtested holdout prediction of daily mutual fund returns on a variety of asset classes

- Using a DL latent factor model to produce better factors than our benchmark: Fama-French's 5-factor model
- Now targetting SHAP explainability publication

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA

Integrated hydrological models into team's DARPA-supported scientific model orchestration portal

Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX

Helped teach graduate-level courses: *Machine Learning for the Geosciences*, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, HPC, & optimization

Key Projects

oppsAlert — Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15000+ points of individual homeowner protest data provided by the City of Austin

Stealth, RegTech — *Seeking PE investment:* Teaming up with a top-proptech operator for a very large opportunity to aid cities achieve permitting compliance via simple pipeline collating publicly available imagery against permitting data

Flood Hazard Assessment — Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.

Texas Disaster Information System (TDIS) — Led development of the core elevation data layer for a web-based spatial data system that supports resilient decision-making at the state and local levels.

Real-Time Inundation Mapping — Developed computationally efficient methods to produce high-resolution (1m) flood inundation maps from National Water Model outputs for emergency response personnel.

Technology Skills

Programming Languages	Python, Bash, PL/pgSQL, Fortran, Perl, C++	Operating Systems	GNU/Linux, macOS, Windows
Libraries & Frameworks	GDAL, CUDA, GeoPandas, RasterIO, NumPy, Plotly, Dash	Database Systems	PostgreSQL, Redis, Supabase, PostGIS
Scientific Software	MODFLOW, HAND, HEC-RAS, MESA	Cloud & Virtualization	AWS, Azure, Docker, Singularity
AI tooling	ChatGPT, Claude, Gemini, VSCode, v0.dev	ML	VAE, Tree-based models
Clustering techniques & metrics	DBSCAN, Hierarchical, Silhouette score, Calinski-Harabasz, Davies-Bouldin	Statistical & ML metrics	R^2, MAPE, RMSE, KL-div, Reconstruction loss

Languages

Spanish	Native / Bilingual Proficiency	French	Professional Working Proficiency

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